

Exercise J02

Q1. Without expanding as far as possible, prove that

$$(a) \begin{vmatrix} a^2 + 2a & 2a + 1 & 1 \\ 20a + 1 & a + 2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (a - 1)^3 \quad (b) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = [(x-y)(y-z)(z-x)(x+y+z)]$$

Ans. -

Sol. -

Q2. If $\begin{bmatrix} x^3 + 1 & x^2 & x \\ y^3 + 1 & y^2 & y \\ z^3 + 1 & z^2 & z \end{bmatrix} = 0$ and x, y, z are all different then, prove that $xyz = -1$.

Ans. -

Sol. -

Q3. Using properties of determinants or otherwise evaluate $\begin{bmatrix} 18 & 40 & 89 \\ 40 & 89 & 198 \\ 89 & 198 & 440 \end{bmatrix}$.

Ans. -18 8. $x = n\pi, n \in \mathbb{I}$

Sol. -

Q4. Prove that $\begin{bmatrix} a - b - c & 2a & 2a \\ 2b & b - c - a & 2b \\ 2c & 2c & c - a - b \end{bmatrix} = (a + b + c)^3$

Ans. -

Sol. -

Q5. Prove that for all values of θ , $\begin{vmatrix} \sin \theta & \cos \theta & \sin 2\theta \\ \sin \left(\theta + \frac{2\pi}{3} \right) & \cos \left(\theta + \frac{2\pi}{3} \right) & \sin \left(2\theta + \frac{4\pi}{3} \right) \\ \sin \left(\theta - \frac{2\pi}{3} \right) & \cos \left(\theta - \frac{2\pi}{3} \right) & \sin \left(2\theta - \frac{4\pi}{3} \right) \end{vmatrix} = 0$

Ans. -

Sol. -

Q6. Prove that $\begin{vmatrix} (\beta + \gamma - \alpha - \delta)^4 & (\beta + \gamma - \alpha - \delta)^2 & 1 \\ (\gamma + \alpha + \beta - \delta)^4 & (\gamma + \alpha + \beta - \delta)^2 & 1 \\ (\alpha + \beta - \gamma - \delta)^4 & (\alpha + \beta - \gamma - \delta)^2 & 1 \end{vmatrix} = -64(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta)$

Ans. -

Sol. Let $a = \beta + \gamma - \delta - \alpha$, $b = \gamma + \alpha - \beta - \delta$

$$c = \alpha + \beta - \gamma - \delta$$

we get
$$\begin{vmatrix} a^4 & a^2 & 1 \\ b^4 & b^2 & 1 \\ c^4 & c^2 & 1 \end{vmatrix}$$

$$R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$$

$$\begin{vmatrix} a^4 - b^4 & a^2 - b^2 & 0 \\ b^4 - c^4 & b^2 - c^2 & 0 \\ c^4 & c^2 & 1 \end{vmatrix}$$

$$= (a^2 - b^2)(b^2 - c^2) \begin{vmatrix} a^2 + b^2 & 1 & 0 \\ b^2 + c^2 & 1 & 0 \\ c^4 & c^2 & 1 \end{vmatrix}$$

$$= (a^2 - b^2)(b^2 - c^2)(a^2 - c^2)$$

$$= (a - b)(a + b)(b - c)(b + c)(a + c)(a - c)$$

$$= 2^6 (\alpha - \beta)(\alpha - \beta)(\alpha - \gamma)(\beta - \gamma)(\alpha - \delta)(\beta - \delta)(\gamma - \delta)$$

Q7. Solve for x the equation

$$\begin{vmatrix} a^2 & a & 1 \\ \sin(n+1)x & 0 \sin nx & \sin(n-1)x \\ \cos(n+1)x & \cos nx & \cos(n-1)x \end{vmatrix} = 0$$

Ans. -

Sol. -

Q8. Solve for x

$$(a) \begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+5 & 5x+8 & 10x+17 \end{vmatrix} = 0. \quad (b) \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0.$$

Ans. (a) $x = -1$ or $x = -2$; (b) $x = 4$

Sol. -

Q9.

If $a + b + c = 0$, solve for x :
$$\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0.$$

Ans. $x = 0$ or $x = \pm \sqrt{\frac{3}{2}(a^2 + b^2 + c^2)}$

Sol. On applying $R_1 \rightarrow R_1 + R_2 + R_3$ and then solving the determinant we get $x = 0$ or $x^2 = a^2 + b^2 + c^2 - ab - bc - ca$.

$$x = \sqrt{\frac{3}{2}(a^2 + b^2 + c^2)}$$

Q10. If $p + q + r = 0$, prove that $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix} = pqr \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$.

Ans. -

Sol. -

Q11. If a, b, c are all different & $\begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$, then prove that, $abc(ab+bc+ca) = a + b + c$.

Ans. -

Sol. -

Q12. Show that, $\begin{vmatrix} a^2 + \lambda & ab & ac \\ ab & b^2 + \lambda & bc \\ ac & bc & c^2 + \lambda \end{vmatrix}$ is divisible by λ^2 and find the other factor.

Ans. $\lambda^2 (a^2 + b^2 + c^2 + \lambda)$

Sol. -

Q13. Evaluate $\begin{vmatrix} \sqrt{13} + \sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} + \sqrt{26} & 5 & \sqrt{10} \\ 3 + \sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$

Ans. $5(3\sqrt{2} - 5\sqrt{3})$

Sol. $\begin{vmatrix} \sqrt{13} + \sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} + \sqrt{26} & 5 & \sqrt{10} \\ 3 + \sqrt{65} & \sqrt{15} & 5 \end{vmatrix} = \begin{bmatrix} \sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} & 5 & \sqrt{10} \\ 3 & \sqrt{15} & 5 \end{bmatrix} + \begin{bmatrix} \sqrt{13} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{26} & 5 & \sqrt{10} \\ \sqrt{65} & \sqrt{15} & 5 \end{bmatrix}$

$$= \sqrt{3}\sqrt{5}\sqrt{5} \begin{bmatrix} 1 & 2 & 1 \\ \sqrt{5} & \sqrt{5} & \sqrt{2} \\ \sqrt{3} & \sqrt{3} & \sqrt{5} \end{bmatrix} + \sqrt{13}\sqrt{5}\sqrt{5} \begin{bmatrix} 1 & 2 & 1 \\ \sqrt{2} & \sqrt{5} & \sqrt{2} \\ \sqrt{5} & \sqrt{3} & \sqrt{5} \end{bmatrix}$$

$$C_1 \rightarrow C_1 - C_2$$

$$= 5\sqrt{3} \begin{bmatrix} -1 & 2 & 1 \\ 0 & \sqrt{5} & \sqrt{2} \\ 0 & \sqrt{3} & \sqrt{5} \end{bmatrix} = 5(3\sqrt{2} - 5\sqrt{3})$$

Q14. Let $\Delta_r = \begin{vmatrix} r-1 & n & 6 \\ (r-1)^2 & 2n^2 & 4n-2 \\ (r-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$. Find the sum $\sum_{r=1}^n \Delta_r$

Ans. 0

Sol. -

Q15. Prove that $\begin{vmatrix} (a_1 - b_1)^2 & (a_1 - b_2)^2 & (a_1 - b_3)^2 \\ (a_2 - b_1)^2 & (a_2 - b_2)^2 & (a_2 - b_3)^2 \\ (a_3 - b_1)^2 & (a_3 - b_2)^2 & (a_3 - b_3)^2 \end{vmatrix} = 2(a_1 - a_2)(a_2 - a_3)(a_3 - a_1)(b_1 - b_2)(b_2 - b_3)(b_3 - b_1)$

Ans. -

Sol. -

Q16. Prove that $\begin{vmatrix} 2 & \alpha + \beta + \gamma + \delta & \alpha\beta + \gamma\delta \\ \alpha + \beta + \gamma + \delta & 2(\alpha + \beta)(\gamma + \delta) & \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) \\ \alpha\beta & \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) & 2\alpha\beta\gamma\delta \end{vmatrix} = 0$.

Ans. -

Sol. -

Q17. Prove that : $\begin{vmatrix} (a-p)^2 & (a-q)^2 & (a-r)^2 \\ (b-p)^2 & (b-q)^2 & (b-r)^2 \\ (c-p)^2 & (c-q)^2 & (c-r)^2 \end{vmatrix} = \begin{vmatrix} (1+ap)^2 & (1+aq)^2 & (1+ar)^2 \\ (1+bp)^2 & (1+bq)^2 & (1+br)^2 \\ (1+cp)^2 & (1+cq)^2 & (1+cr)^2 \end{vmatrix}$

Ans. -

Sol. -

Q18. If $S_r = \alpha^r + \beta^r + \gamma^r$ then show that $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix} = (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2$.

Ans. -

Sol. $\begin{vmatrix} 3 & \alpha + \beta + \gamma & \alpha^2 + \beta^2 + \gamma^2 \\ \alpha + \beta + \gamma & \alpha^2 + \beta^2 + \gamma^2 & \alpha^3 + \beta^3 + \gamma^3 \\ \alpha^2 + \beta^2 + \gamma^2 & \alpha^3 + \beta^3 + \gamma^3 & \alpha^4 + \beta^4 + \gamma^4 \end{vmatrix}$
 $= \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix}$
 $= (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2$

Q19. Find the real values of r for which the following system of linear equations has a non-trivial solution. Also find the non-trivial solutions :

$$2rx - 2y + 3z = 0$$

$$x + ry + 2z = 0$$

$$2x + rz = 0$$

Ans. $r = 2$; $x = k$; $y = k/2$; $z = -k$ where $k \in \mathbb{R} \setminus \{0\}$

Sol. -

Q20. Solve the following using Cramer's rule and state whether consistent or not.

(a) $x + y + z - 6 = 0$

(b) $x + 2y + z = 1$

(c) $7x - 7y + 5z = 3$

$2x + y - z - 1 = 0$

$3x + y + z = 6$

$3x + y + 5z = 7$

$x + y - 2z + 3 = 0$

$x + 2y = 0$

$2x + 3y + 5z = 5$

Ans. (a) $x = 1$, $y = 2$, $z = 3$; consistent (b) $x = 2$, $y = 1$, $z = 1$; consistent, (c) inconsistent

Sol.

$$(a) D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & -2 \end{vmatrix} = 3$$

$$\Rightarrow D_1 = \begin{vmatrix} 6 & 1 & 1 \\ 1 & 1 & -1 \\ -3 & 1 & -2 \end{vmatrix}$$

$$D_2 = \begin{vmatrix} 1 & 6 & 1 \\ 2 & 1 & -1 \\ 1 & -3 & -2 \end{vmatrix} = 5$$

Similarly $D_3 = 9$

so consistent having $x = 4$, $y = 2$, $z = 3$

$$(c) D = \begin{vmatrix} 7 & -7 & 5 \\ 3 & 1 & 5 \\ 2 & 3 & 5 \end{vmatrix} = 0$$

$$D_1 = \begin{vmatrix} 1 & 6 & 1 \\ 2 & 1 & -1 \\ 1 & -3 & -2 \end{vmatrix} = 24$$

Inconsistent system

Q21. For what value of K do the following system of equations possess a non trivial (i.e. not all zero) solution over the set of rationals $\mathbb{Q}$?

$$x + Ky + 3z = 0, 3x + Ky - 2z = 0, 2x + 3y - 4z = 0.$$

For that value of K , find all the solutions of the system.

Ans. $K = \frac{33}{2}, x : y : z = -\frac{15}{2} : 1 : -3$

Sol. -

Q22. The system of equations

$$\alpha x + y + z = \alpha - 1$$

$$x + \alpha y + z = \alpha - 1$$

$$x + y + \alpha z = \alpha - 1 \quad \text{has no solution. Find } \alpha.$$

Ans. -

Sol. -

Q23. Given $x = cy + bz$; $y = az + cx$; $z = bx + ay$ where x, y, z are not all zero, prove that $a^2 + b^2 + c^2 + 2abc = 1$.

Ans. -

Sol. -

Q24. For what values of p , the equations : $x + y + z = 1$; $x + 2y + 4z = p$ & $x + 4y + 10z = p^2$ have a solution? Solve them completely in each case.

Ans. $x = 1 + 2K, y = ? 3K, z = K$, when $p = 1$; $x = 2K, y = 1 + 3K, z = K$ when $p = 2$; where $K \in \mathbb{R}$

Sol. -

Q25. If the following system of equations $(a - t)x + by + cz = 0$, $bx + (c - t)y + az = 0$ and $cx + ay + (b - t)z = 0$ has non-trivial solutions for different values of t , then show that we can express product of these values of t in the form of determinant.

Ans. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Sol. -

Q26. The set of equations $x - y + 3z = 2$, $2x - y + z = 4$, $x - 2y + \alpha z = 3$ has

(A) unique solution only for $\alpha = 0$ (B) unique solution for $\alpha \neq 8$

(C) infinite number of solutions for $\alpha = 8$ (D) no solution for $\alpha = 8$

Ans. D, B

Sol. $D = \alpha - 8 \Rightarrow (B)$, If $\alpha = 8, D_x = 2\alpha - 14 \Rightarrow$ no solution

Q27. If A_{11}, A_{12}, A_{13} are the cofactors of the elements of the first row of the determinant $A = (a_{ij})$ and Δ is the value of the determinant then

(A) $a_{21} A_{11} + a_{22} A_{12} + a_{23} A_{13} = 0$ (B) $a_{11} A_{11} + a_{12} A_{12} + a_{13} A_{13} = \Delta$

(C) $a_{11} A_{11} - a_{12} A_{12} + a_{13} A_{13} = \Delta$ (D) $a_{31} A_{31} + a_{32} A_{12} + a_{33} A_{13} = 0$

Ans. A, B, D

Sol. -

Q28. The determinant $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$ if

(A) a, b, c are in A.P. (B) a, b, c are in G.P. (C) α is a roots of $ax^2 + bx + c = 0$

(D) $(x - \alpha)$ is a factor of $ax^2 + 2bx + c = 0$

Ans. D, B

Sol. -

Q29. The parameter, on which the value of $\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$ depend upon, is

(A) a (B) p (C) d (D) x

Ans. D, A, C

Sol. -

Q30. **Column-I**

Column-II

(A) Let $|A| = |a_{ij}|_{3 \times 3} \neq 0$. Each element a_{ij} is multiplied by k^{i-j} . Let $|B|$ the resulting Determinant, where $k_1|A| + k_2|B| = 0$. Then $k_1 + k_2 =$

(P) 0

(B) The maximum value of a third order determinant each of its entries are ± 1 equals

(Q) 4

(C) If $\begin{vmatrix} 1 & \cos \alpha & \cos \beta \\ \cos \alpha & 1 & \cos \gamma \\ \cos \beta & \cos \gamma & 1 \end{vmatrix}$
 $= \begin{vmatrix} 1 & \cos \alpha & \cos \beta \\ \cos \alpha & 0 & \cos \gamma \\ \cos \beta & \cos \gamma & 0 \end{vmatrix}$ then $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma =$

(R) 1

(D) $\begin{vmatrix} x^2 + x & x + 1 & x - 2 \\ 2x^2 + 3x - 1 & 3x & 3x - 3 \\ x^2 + 2x + 3 & 2x - 1 & 2x - 1 \end{vmatrix}$ (S) 2

$= Ax + B$ where A and B are determinants of order 3.

Then $A + 2B =$

- (A) $(A) - P(B) - Q(C) - R(D) P$ (B) $(A) - Q(B) - P(C) - R(D) - S$
(C) $(A) - R(B) - S(C) - Q(D) P$ (D) $(A) - S(B) - Q(C) - P(D) R$

Ans. A

Sol. -